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Using the sample data given in Exhibit 10, make a recommendation for how many units of each style Wally Obermeyer should order during the initial phase of production. Assume that all ten styles in the sample problem are made in Hong Kong (at this point, ignore the minimum production constraints), and that Obermeyer's initial production commitment must be at least 10,000 units. Ignore price differences among styles in your analysis. This is a group assignment (groups of 3 + one group of 4 or two groups of 2). Please have only one group member upload your answers.

Problem

We must commit at least 10,000 units of the 10 sample parkas before the Las Vegas trade show, with the remaining 10,000 units produced after the show.

Demand model

Demand for each style i is normal:

- μ_i
- σ_i = 2 $\times$ standard deviation

Economics

Each parka sold earns 24% of the wholesale price, and each unsold parka loses 8%. The critical ratio is

$$\frac{C_u}{C_u + C_o} = \frac{0.24}{0.24 + 0.08} = 0.75$$

Approach: equal risk across styles

The units made now should be the ones we are most confident of selling.

$$Q_i = \mu_i + z\sigma_i, \quad \sum_i Q_i = 10000$$

$$\mu + z\sigma = 10000$$

$$20001 + z \cdot 9428 = 10000$$
$$z = \frac{10000 - 20001}{9428}$$
$$= -1.0608$$

This gives $z = -1.0608$.

Style	Q^*
Gail	1278.7
Isis	1477.72
Entice	1692.55
Assault	2983.65
Teri	1613.96
Electra	2694.99
Stephanie	1819.87
Educed	4767.03
Anita	4708.38
Delphne	3323.24
Total	26360.09

Style	**New** Q^*
Gail	605.42
Isis	356.74
Entice	831.85
Assault	1803.67
Teri	291.69
Electra	1292.89
Stephanie	1.31
Educed	2837.42
Anita	1074.73
Delphne	904.28
Total	10000

Recommendation

Wally should commit the quantities in the table above (total 10,000 units), rounded to whole units. Stephanie comes out at roughly one unit, so its production should be deferred entirely until after Las Vegas.

Interpretation

- High-consensus styles (Seduced, Assault, Gail, Entice) are produced mostly up front: their demand is predictable, so early production carries little risk.
- High-disagreement styles (Stephanie, Anita, Teri, Isis) are cut back hardest: their demand is uncertain, so most of their production is postponed until the trade show provides better information.
- The approach turns the committee's disagreement into a production-sequencing rule: make the safe units now, wait on the risky ones. The flexible post-show capacity is kept for the styles where information is worth the most.